GNSS Principles Experiment Report

Week 3, Lab 2, [Lab 2 Topic], [Date]

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1 Task 1: Ephemeris Parameters of Tracked Satellites

1.1 Ephemeris Parameter Values

The ephemeris parameters for four tracked satellites, obtained from self-collected data, are presented in Table 1. These parameters are critical for determining the precise positions and clock corrections of the satellites, enabling accurate GPS positioning.

Table 1: Ephemeris Parameters for Tracked Satellites

Parameter	Satellite 1	Satellite 2	Satellite 3	Satellite 4
C_ic	7.45058059692383e-08	9.68575477600098e-08	-5.21540641784668e-08	1.06170773506165e-07
omega_0	2.18178063205793	1.16461010593126	-2.0594276074262	1.09480371771214
C_is	3.53902578353882e-08	1.67638063430786e-07	8.5681676864624e-08	5.58793544769287e-08
i_0	0.96438457387131	0.959117717685609	0.988370568458548	0.943796711229907
C_rc	187.84375	277	362.25	272.28125
omega	-1.22575689248322	-1.06644426662146	1.12826294804053	-2.8506971369439
omegaDot	-7.54459997715809e-09	-7.93033033008986e-09	-8.30677458193254e-09	-8.45892377670007e-09
IODE_sf3	75	56	1	4
iDot	-5.382367054335e-10	-1.56077929843689e-10	1.2500520696863e-10	-1.22147945095061e-10
weekNumber	1333	1333	1333	1333
accuracy	0	0	0	0
health	0	0	0	0
T_GD	-1.0710209608078e-08	-1.11758708953857e-08	1.39698386192322e-09	-7.91624188423157e-09
IODC	233	232	3	495
t_oc	115200	115200	115184	115200
a_f2	0	0	0	0
a_f1	-1.92130755749531e-11	-1.13686837721616e-12	4.66116034658626e-12	6.3664629124105e-12
a_f0	0.000386329367756844	-9.96338203549385e-05	0.000673974864184856	0.000611822120845318
IODE_sf2	75	56	1	4
C_rs	6.28125	12.15625	-0.4375	13.03125
deltan	4.14588697854815e-09	4.78841374236691e-09	4.27553523606133e-09	5.3002207754699e-09
M_0	-3.0583215673691	-2.55216136093054	0.00163461784839459	-0.404358350074052
C_uc	4.71249222755432e-07	6.48200511932373e-07	-1.47148966789246e-07	4.88013029098511e-07
e	0.0131110963411629	0.0132631108863279	0.0058371463092044	0.00543699401896447
C_us	1.00079923868179e-05	5.29550015926361e-06	1.87195837497711e-06	5.13531267642975e-06
sqrtA	5153.64450645447	5153.65659141541	5153.56258964539	5153.70109176636
t_oe	115200	115200	115184	115200
TOW	113202	113202	113202	113202

1.2 Meaning and Function of Each Parameter

C_ic Amplitude of the cosine harmonic correction term to the angle of inclination, used to adjust the inclination for orbital perturbations.

omega_0 Longitude of the ascending node of the orbit plane at the weekly epoch, defining the orbital plane's orientation relative to the Earth's equator.

C_is Amplitude of the sine harmonic correction term to the angle of inclination, providing a sinusoidal adjustment to the inclination.

i_0 Inclination angle at the reference time, the angle between the orbital plane and the equatorial plane.

C_rc Amplitude of the cosine harmonic correction term to the orbit radius, correcting the radial distance of the satellite.

omega Argument of perigee, the angle from the ascending node to the perigee, indicating the satellite's position within its orbit.

omegaDot Rate of change of the longitude of the ascending node, accounting for the precession of the orbital plane.

IODE_sf3 Issue of Data, Ephemeris for subframe 3, ensuring the ephemeris data's consistency.

iDot Rate of change of the inclination angle, representing the temporal variation in the orbit's inclination.

weekNumber GPS week number, indicating the week since the GPS time epoch (January 6, 1980).

accuracy User Range Accuracy (URA) index, a measure of the satellite's expected ranging accuracy.

health Satellite health status, where 0 indicates the satellite is operational.

T_GD Group delay differential, a correction for single-frequency receivers to account for signal delays.

IODC Issue of Data, Clock, an identifier for the clock correction data set.

t_oc Reference time for clock data.

a_f2 Satellite clock drift rate (second-order term).

a_f1 Satellite clock drift (first-order term).

a_f0 Satellite clock bias (constant term).

IODE_sf2 Issue of Data, Ephemeris for subframe 2.

C_rs Amplitude of the sine harmonic correction term to the orbit radius.

deltan Mean motion difference correction.

M_0 Mean anomaly at the reference time.

C_uc Amplitude of the cosine harmonic correction term to the argument of latitude.

e Eccentricity of the orbit.

C_us Amplitude of the sine harmonic correction term to the argument of latitude.

sqrtA Square root of the semi-major axis.

t_oe Reference time of the ephemeris.

TOW Time of Week when the data was collected.

2 Task 2: Satellite Positioning and Clock Corrections

2.1 Positioning Results and Clock Corrections

The positions and clock correction parameters for four tracked satellites are provided in Table 2. The positions are given in the Earth-Centered, Earth-Fixed (ECEF) coordinate system (in meters), and the clock corrections are in seconds. These measurements were obtained using a Kalman Filter (KF) process.

Table 2: Positioning Results and Clock Corrections for Tracked Satellites

Parameter	Satellite 1	Satellite 2	Satellite 3	Satellite 4
X (m)	-6894914.7127	-5462176.5460	-19517395.8741	-10726860.9577
Y (m)	14047248.6733	21346707.8335	6933283.0231	22605748.1180
Z (m)	21881194.1621	15425560.1574	16386558.4811	8575114.1861
Clock Correction (s)	0.0003863723	-0.0000996116	0.0006739680	0.0006118253

Notes:

- The position coordinates are precise to several decimal places.
- Clock corrections account for time discrepancies between the receiver and the satellites.

2.2 Why At Least Four Satellites Are Required

Accurate GPS positioning requires solving for four unknowns:

- Receiver's X-coordinate (ECEF)
- Receiver's Y-coordinate (ECEF)
- Receiver's Z-coordinate (ECEF)
- Receiver's clock error (δt)

Each satellite provides a pseudorange measurement (ρ_i), modeled as:

$$\rho_i = \sqrt{(X_i - X)^2 + (Y_i - Y)^2 + (Z_i - Z)^2} + c\,\delta t + \epsilon_i$$

where:

- (X_i, Y_i, Z_i): Known satellite coordinates.
- (X, Y, Z): Unknown receiver coordinates.
- c: Speed of light.

- δt: Receiver clock error.
- ϵ_i: Measurement noise.

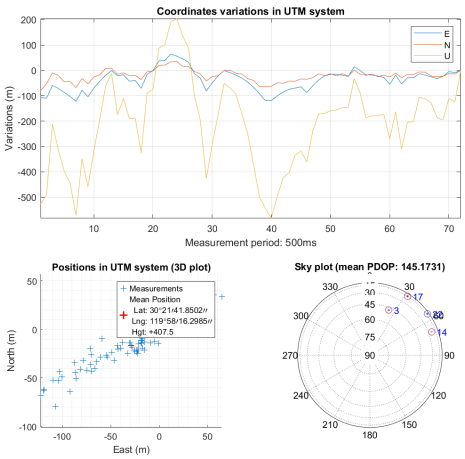
Since there are four unknowns, at least four satellites (i.e., four equations) are needed to uniquely determine the receiver's position and clock error.

3 Task 3: Positioning Results, DOP Variation, and Tropospheric Correction Analysis

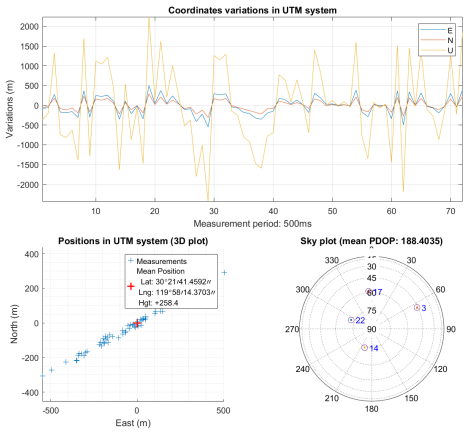
3.1 Positioning Results

The positioning results are visualized using:

- **Trajectory Maps** (for Kalman Filter and Least Squares)
- **Coordinate Variation Plots** (East, North, Up over time)



(a) Kalman Filter



(b) Least Squares

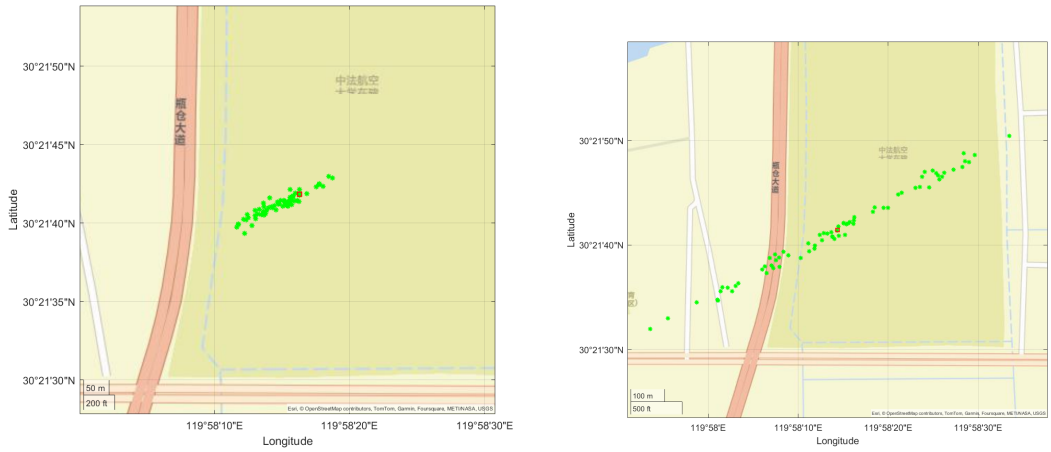
Figure 1: Trajectory maps with tropospheric correction.

3.2 DOP Variation

The PDOP plot in Figure 3 illustrates how satellite geometry affects positioning accuracy over time.

3.3 Impact of `initSettings.useTropCorr`

Enabling `initSettings.useTropCorr` applies a tropospheric correction, which improves vertical (Up) accuracy. Differences can be observed in UTM height values (e.g., +407 vs. +258) and in the mean PDOP values (from 145.17 to 188.40).



(a) Kalman Filter (b) Least Squares

Figure 2: Coordinate variation over time with tropospheric correction.

3.4 Summary

The Kalman Filter produces smoother position estimates than the least squares method, and tropospheric correction significantly improves vertical accuracy. The variation in DOP further emphasizes the role of satellite geometry in positioning performance.

4 Task 4: Effect of Changing the Initial Value on the Kalman Filter

4.1 Results and Error Message

After using a least squares estimation to set the initial values, no positioning results were obtained. The following error was encountered:

```
Index exceeds the number of array elements. Index must not exceed 4.
Error in postNavigationKalman (line 200)
            X([1 3 5]) = [xyzdt(1), xyzdt(2), xyzdt(5)];
```

5 Task 5: Adjusting Parameters and Testing Positioning Performance

5.1 Experiment

The following parameter values were tested:

- Sg = 2e3

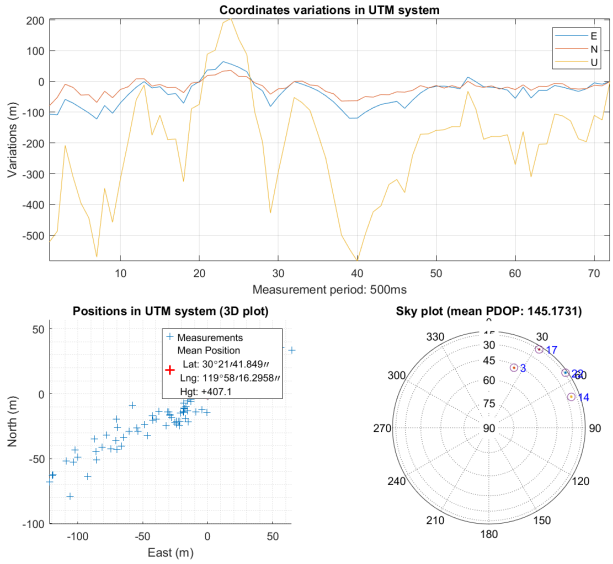


Figure 3: PDOP variation over time.

```
%% Kalman Initialization
syms x Vx y Vy z Vz
syms b d
T = settings.navSolPeriod/1000 ;

Xstate = [x Vx y Vy z Vz b d].';

f = [x+T*Vx;
    Vx;
    y+T*Vy;
    Vy;
    z+T*Vz;
    Vz;
    b+T*d
    d];
%% Try to adjust these parameters to obtain a better Q and P
Sf = 1e5;
Sg = 100;
Svx = 0.01; Svy = 0.01; Svz = 0.01;
```

Figure 4: Results from parameter adjustment experiment.

- Svx = 0.8
- Svy = 0.1
- Svz = 0.2

6 Task 6: Correcting Ionospheric Parameters

This section is reserved for discussing the correction of ionospheric parameters.

function iono$_d elay = ionos(TOW, lat, lon, el, az, eph)$

c = 299792458;

if isfield(eph, 'alpha') —— isfield(eph, 'beta') alpha = [2.5e-08, 1.0e-08, -1.0e-07, 0]; beta = [1.2e+05, 0, -1.6e+05, 0]; else alpha = eph.alpha; beta = eph.beta; end

$\text{lat}_s c = lat/180; lon_s c = lon/180; el_s c = el/180;$

psi = 0.0137 / ($\text{el}_s c + 0.11$) − 0.022;

$\text{lat}_i = lat_s c + psi * cos(az); lat_i = min(0.416, max(-0.416, lat_i));$

$\text{lon}_i = lon_s c + (psi * sin(az))/cos(lat_i * pi);$

$\text{lat}_m = lat_i + 0.064 * cos((lon_i - 1.617) * pi);$

t = 43200 * $\text{lon}_i + TOW; t = mod(t, 86400);$

F = 1.0 + 16.0 * (0.53 - $\text{el}_s c)^3$;

AMP = alpha(1) + alpha(2)*$\text{lat}_m + alpha(3) * lat_m^2 + alpha(4) * lat_m^3; if AMP < 0 AMP = 0; end$

PER = beta(1) + beta(2)*$\text{lat}_m + beta(3) * lat_m^2 + beta(4) * lat_m^3; if PER < 72000 PER = 72000; end$

X = 2 * pi * (t - 50400) / PER;

if abs(X) ¡ 1.57 $\text{delay}_s ec = F * (5e-9 + AMP * (1-(X^2)/2+(X^4)/24)); else delay_s ec = F * 5e-9; end$

$\text{iono}_d elay = c * delay_s ec; end$